

Kushagramati Analytics Pvt Ltd (K Mati)

Detailed Proposal and Project Report

Namma Clinic Digital Health and Operations Platform

A proposal for strengthening clinic operations, public health monitoring, citizen service, analytics, and ABDM-ready digital health records across Namma Clinics under GBA / BBMP.

Prepared for

Sri Venkata Rao Chalapathi, IAS
Special Commissioner
Greater Bengaluru Authority / BBMP Health Department
Karnataka

Note: Name/designation and official address should be verified from the latest official directory before dispatch.

Prepared by

Kushagramati Analytics Pvt Ltd (K Mati)
No. 16, 2nd Floor, 17th Cross, M C Road, Near Maruthi Mandir, Vijayanagar, Bengaluru - 560040
Email: sales@kmati.in | Website: www.kmati.in
Date: June 2026 | Version: 1.0

Commercial model in this report is indicative. Final pricing depends on confirmed clinic count, existing hardware, hosting preference, integration scope, and procurement conditions.

Executive Snapshot

Core proposal

K Mati proposes to design, implement, and operate a unified Namma Clinic Digital Health and Operations Platform covering patient registration, queue management, doctor console, EMR-lite, diagnostics, pharmacy and medicine inventory, referrals, telemedicine support, public-health reporting, and a GBA command dashboard.

Decision Item	Recommended Position
Primary objective	Improve citizen access, reduce paper registers, standardize clinic workflows, improve referral tracking, prevent medicine stock-outs, and provide near-real-time public-health visibility.
Recommended start	4-week discovery and DPR, followed by a 20-clinic pilot across different zones and patient-load conditions.
Recommended rollout	9 to 12 months for citywide rollout after pilot approval, with phase-wise deployment across GBA / BBMP zones or new city corporations.
Budget planning baseline	Recommended full rollout scenario for 200 clinics: approximately Rs. 12.9 crore year-one project cost plus Rs. 5.6 crore annual recurring operations from year two.
Pilot budget	20-clinic pilot: approximately Rs. 1.65 crore including discovery hardening, configuration, training, devices/peripherals, hosting, first-year support, and contingency.
Strategic outcome	A measurable, auditable, ABDM-ready primary-care operations platform for urban health governance.

Why this matters now

- **For citizens:** faster registration, better follow-up, fewer lost prescriptions/reports, easier referral continuity, and clearer access to medicines and screenings.
- **For clinic staff:** less duplicate paperwork, one screen for visit notes and prescriptions, easier inventory tracking, and faster reporting.
- **For GBA / BBMP:** zone-wise operational visibility, disease trend monitoring, medicine stock-out control, referral closures, and measurable public service improvement.
- **For K Mati:** a flagship GovTech and HealthTech reference project aligned to its AI, analytics, data engineering, healthcare, and platform-modernization strategy.

Document Index

This index is designed for proposal submission. In Microsoft Word, use References -> Update Table after opening the file to generate page numbers if needed.

1. 1. Executive Summary
2. 2. Background and Need for Digital Namma Clinic Platform
3. 3. K Mati Value Proposition and Fit
4. 4. Project Vision, Objectives, and Outcomes
5. 5. Proposed Solution Overview
6. 6. Detailed Functional Scope
7. 7. Patient, Clinic, and Officer Workflows
8. 8. Data Platform, AI, and Analytics Strategy
9. 9. Interoperability with ABDM, eHospital, and Government Systems
10. 10. Architecture and Technical Design
11. 11. Security, Privacy, Compliance, and Data Governance
12. 12. Benefits for Public, Clinics, Government, and K Mati
13. 13. Implementation Plan and Timeline
14. 14. Budget Plan and Commercial Options
15. 15. Pilot Plan, Success Metrics, and Acceptance Criteria
16. 16. Operations, Training, Helpdesk, and SLA
17. 17. Risk Register and Mitigation Plan
18. 18. Governance Model and Decision Structure
19. 19. Procurement Strategy and Funding Models
20. 20. Appendices: Modules, Roles, Dashboards, Data Dictionary, Cover Letter, and References

1. Executive Summary

Kushagramati Analytics Pvt Ltd, operating as K Mati, proposes a comprehensive digital health and operations platform for Namma Clinics serving Bengaluru citizens under the Greater Bengaluru Authority / BBMP health administration. The proposed platform is designed as a practical, clinic-ready system for public primary care: easy enough for daily frontline use, strong enough for municipal health governance, and modular enough to integrate with India's digital health ecosystem over time.

The proposal is not only a software implementation. It is a service-delivery modernization program. The intent is to improve patient registration, clinic workflow, doctor documentation, pharmacy stock visibility, referral continuity, disease surveillance, preventive health monitoring, and officer-level dashboards. The platform will help the administration see what is happening across clinics every day rather than relying only on delayed manual reports.

The platform is designed around five operating principles:

- Clinic-first: registration, triage, consultation, prescription, lab, pharmacy, and referral workflows should be completed quickly with minimal typing burden.
- Citizen-first: patients should receive clear visit summaries, prescriptions, referral instructions, and follow-up reminders in Kannada and English.
- ABDM-ready: the solution should support ABHA-enabled patient identification and future consent-based digital health record exchange wherever approved by policy.
- Data-governed: all personal and health data should be protected with role-based access, audit logs, encryption, and India-hosted log retention.
- Outcome-led: dashboards should track patient service, medicine stock-outs, NCD screenings, referrals, public-health patterns, and clinic performance indicators.

Recommended decision

Approve a 20-clinic pilot covering high-volume, medium-volume, and low-volume clinics across representative GBA / BBMP zones. The pilot should prove daily workflow adoption, patient throughput improvement, inventory visibility, referral tracking, and dashboard usefulness before a full citywide rollout.

2. Background and Need for Digital Namma Clinic Platform

Namma Clinics are intended to strengthen accessible urban primary care. Their value is highest when citizens can reach basic services close to their residence or workplace, when doctors and nurses can work without administrative overload, and when public-health leaders can see disease, medicine, and service patterns at ward, zone, and city level.

The current urban health context makes a unified digital operating layer important for five reasons:

- **High citizen expectation:** Bengaluru citizens increasingly expect fast, transparent, digital public services comparable to private-sector service experiences.
- **Primary care continuity:** Common conditions such as hypertension, diabetes, fever, respiratory illness, maternal and child health follow-up, and immunization-related advice need repeat monitoring.
- **Operational control:** Clinics need predictable medicine availability, test-result capture, staff attendance visibility, and escalation mechanisms.
- **Public-health intelligence:** Municipal health leadership needs early signals for disease clusters, NCD burden, referrals, medicine usage, and service gaps.
- **Administrative restructuring:** Under GBA / BBMP transition and zonal or corporation-level governance, health dashboards must support both local control and citywide oversight.

A clinic-level digital platform should not become a heavy hospital ERP. It should be lighter, faster, and more purpose-built for primary care. The design should prioritize OPD registration, queue, triage, consultation, prescription, lab, pharmacy, referral, and dashboards. More complex hospital functions can remain with existing hospital systems and be integrated later through APIs or referral links.

3. K Mati Value Proposition and Fit

K Mati is well positioned to present this proposal because its internal materials describe a move from generic analytics toward AI automation, GovTech, HealthTech, EduTech, and data modernization. The company strategy also identifies healthcare exposure, ABDM, hospital automation, analytics dashboards, and data engineering as reusable capabilities for immediate market offers.

K Mati capability	Relevance to Namma Clinic platform
Data engineering and ETL	Build structured data flows from clinics to central dashboards, validate patient, visit, pharmacy, referral, and inventory records, and automate daily reporting.
Cloud and data modernization	Deploy scalable, secure, modular platform components that can run on government cloud, India-hosted cloud, or state data-center infrastructure.
AI / ML and analytics	Create early warning indicators, NCD risk cohorts, stock-out prediction, patient follow-up prioritization, and officer-level decision dashboards.
Application modernization	Build workflow applications for doctors, nurses, pharmacists, lab staff, and supervisors using modern APIs and mobile/web interfaces.
Healthcare domain orientation	Map clinical and operational requirements into a simple EMR-lite and public-health reporting system without overcomplicating clinic workflows.

Project management experience	Run discovery, pilot, training, rollout, support, and governance through milestone-based program management.
-------------------------------	--

The right positioning for K Mati is not: "we will sell software." The right positioning is: "we will help GBA / BBMP make Namma Clinics measurable, citizen-friendly, digital, and governance-ready." This message is more persuasive for government because it connects technology to public outcomes.

4. Project Vision, Objectives, and Outcomes

Project vision

To create a unified, secure, multilingual, ABDM-ready digital health operations platform that strengthens Namma Clinic service delivery, improves public access, supports doctors and frontline staff, and gives GBA / BBMP leaders near-real-time visibility across urban primary health services.

Project objectives

- Digitize patient registration, queue, triage, consultation, prescription, diagnostic, pharmacy, referral, and follow-up workflows.
- Create a patient-centric visit record that can support repeat care and continuity at Namma Clinics.
- Support ABHA-enabled identity and future ABDM integration where policy, consent, and infrastructure permit.
- Reduce manual registers and duplicate data entry for clinic staff.
- Provide live dashboards for clinic-in-charge, zone health officer, health commissioner, and GBA leadership.
- Create public-health analytics for NCD screening, disease trends, medicine utilization, referrals, stock-outs, and service gaps.
- Enable a scalable rollout across all Namma Clinics using standardized templates and configurable workflows.

Target outcomes

Outcome area	Target outcome after rollout
Citizen service	Reduced waiting time, faster revisit retrieval, clearer prescriptions, digital or printed visit summaries, and follow-up reminders.
Clinical quality	Consistent vitals capture, problem list, prescription records, referral notes, and preventive-care prompts.
Inventory	Lower stock-out risk through medicine ledger, expiry alerts, reorder thresholds, and zone-level visibility.
Public health	Faster visibility into fever clusters, respiratory symptoms, vector-borne disease flags, NCD burden, maternal/child referrals, and service utilization.
Governance	Daily dashboard for clinic reporting status, footfall, staffing, medicine stock, referral pendency, and SLA breaches.
Financial control	Better procurement planning, reduced wastage due to expiry, and evidence-based medicine allocation.

5. Proposed Solution Overview

The proposed solution is a modular platform with clinic workflow applications, central data services, analytics dashboards, integration services, security governance, training, and managed support. It can be implemented in a phased manner so that critical modules go live first and advanced analytics are added after adoption stabilizes.

Solution layer	Components
Clinic application layer	Reception and registration, queue/token management, triage, doctor console, lab desk, pharmacy/store, referral desk, clinic supervisor dashboard.

Citizen interaction layer	SMS/WhatsApp/email notifications where permitted, printed visit summary, follow-up reminders, referral instructions, health education messages.
Central services layer	Patient master, clinic master, staff master, medicine master, diagnosis and symptom catalogue, referral directory, consent and audit service.
Analytics layer	Operational dashboard, public-health dashboard, inventory dashboard, disease surveillance, NCD cohorts, referral tracking, AI-driven alerts.
Integration layer	ABHA/ABDM readiness, eHospital-compatible referral exchange, state health reporting exports, lab/pharmacy interfaces where available.
Security layer	Role-based access, encryption, audit logs, incident reporting workflow, backup, disaster recovery, and cyber monitoring.

High-level clinic journey

Step	Digital workflow	Main benefit
1. Patient arrival	Reception searches existing record or creates registration; ABHA-assisted registration is supported where available.	Less manual writing and faster queue creation.
2. Triage	Nurse captures vitals, symptoms, and risk flags.	Doctor gets structured information before consultation.
3. Doctor consultation	Doctor records diagnosis, advice, prescription, tests, and referral if needed.	Consistent medical documentation and visit summary.
4. Lab / diagnostic	Lab technician captures ordered tests and results.	Results linked to patient visit and dashboard.
5. Pharmacy	Medicine issued against prescription; stock automatically updates.	Reduced medicine leakage, wastage, and stock-outs.
6. Referral / follow-up	Referral generated with reason, destination, and follow-up date.	Better continuity and closure tracking.
7. Reporting	Daily data flows to zone and city dashboards.	Officers act on exceptions rather than waiting for manual reports.

6. Detailed Functional Scope

6.1 Patient Registration and Queue Management

- Create patient profile with name, age, gender, phone, address, ward/area, optional ABHA number, and guardian details when needed.
- Search existing patient using mobile number, patient ID, ABHA, or demographic details.
- Generate daily token and route patient to triage, doctor, lab, pharmacy, or referral desk.
- Support walk-in, follow-up, camp, screening, and referred patient categories.
- Print or SMS token if required; allow screen-based queue display in high-volume clinics.
- Capture service type: general OPD, fever clinic, NCD screening, antenatal advice, child health, immunization advice, follow-up, or referral closure.

6.2 Triage and Nursing Workflow

- Record vitals: temperature, pulse, blood pressure, SpO2, height, weight, BMI, blood sugar where test/device is available.
- Capture chief complaint and symptom duration with structured quick-pick options.
- Flag emergency warning signs and route to doctor or referral immediately.

- Capture basic screening for hypertension, diabetes, anemia suspicion, fever, cough, respiratory distress, pregnancy status, elderly risk, and child danger signs.
- Generate nurse queue and doctor-ready summary.

6.3 Doctor Console and EMR-lite

- View patient history, previous visits, allergies/flags, vital signs, current complaints, and pending tests.
- Record diagnosis, clinical notes, advice, prescription, investigation orders, referral decision, and follow-up date.
- Use templates for common primary-care conditions such as fever, cough, hypertension, diabetes, skin conditions, abdominal pain, antenatal advice, and child health.
- Provide clinical documentation that is useful but not burdensome; avoid heavy tertiary-hospital EMR screens.
- Generate printable and digital visit summary with prescription and advice.

6.4 Pharmacy and Medicine Inventory

- Maintain medicine master with generic name, strength, unit, batch number, expiry date, supplier, and reorder level.
- Issue medicines against doctor prescription and update stock automatically.
- Track stock received, stock issued, closing stock, stock transfer, damaged/expired stock, and near-expiry alerts.
- Alert clinic and zone officers for low stock, no stock, fast-moving medicines, and expiry risk.
- Generate procurement planning dashboard using consumption trends.

6.5 Diagnostics and Lab Workflow

- Order and record basic tests such as blood sugar, hemoglobin, urine tests, malaria/dengue rapid test where applicable, and other clinic-approved diagnostics.
- Capture sample collected, result entered, result verified, and result delivered status.
- Link lab result to the patient visit, prescription, referral, and public-health dashboard.
- Allow printout of results and digital delivery where permitted.

6.6 Referral and Follow-up Management

- Generate referral note to UPHC, secondary hospital, maternity hospital, specialist clinic, emergency facility, or public-health program.
- Capture reason for referral, urgency, destination facility, expected date, and follow-up status.
- Track open referrals, closed referrals, delayed referrals, and missed follow-ups.
- Send patient reminder via SMS/WhatsApp where allowed.
- Provide zone-level referral pendency dashboard.

6.7 Telemedicine Support

- Schedule teleconsultation with higher facility or specialist where approved by GBA / state health protocol.
- Capture teleconsult notes and advice in the visit record.
- Enable doctor-to-doctor consult notes without exposing patient data beyond authorized channels.
- Support queue-based telemedicine for select conditions and public-health camps.

6.8 Public Health and Disease Surveillance

- Capture symptoms and diagnoses in structured categories that support public-health analysis.
- Detect abnormal increases in fever, respiratory illness, gastroenteritis, dengue-like symptoms, or other notified disease flags.
- Generate ward/zone heatmaps, trend lines, and alerts for officer review.
- Track NCD screening burden, high-risk cohorts, follow-up due lists, and referrals.
- Support export to approved government reporting formats.

6.9 Citizen Communication and Grievance Linkage

- Send follow-up reminders, referral instructions, medicine availability updates, and health education content in Kannada and English.
- Allow simple citizen feedback after visit, where policy permits.
- Route grievances or service feedback to clinic supervisor / zone health officer dashboard.

- Support AI chatbot in a later phase for clinic timings, services, documents needed, follow-up instructions, and nearest clinic guidance.

7. Patient, Clinic, and Officer Workflows

The platform should be designed around the daily reality of clinics. A doctor or nurse cannot spend many minutes on data entry per patient. The recommended workflow uses pre-filled templates, quick selections, repeat-visit retrieval, and automatic dashboard generation.

7.1 Standard OPD workflow

Role	Actions in platform	Time target
Reception / Data Entry	Search/create patient, create visit, issue token, route to triage or doctor.	60 to 120 seconds for repeat patient; 2 to 4 minutes for new patient.
Nurse / Triage	Capture vitals and chief complaint; mark danger signs; route to doctor.	2 to 5 minutes.
Doctor	Review triage, record diagnosis/advice/prescription/referral, close visit.	3 to 6 minutes for common OPD case.
Lab Technician	Record test order, sample, result, verification, and delivery.	Based on test type.
Pharmacy	Issue medicines and update stock.	1 to 3 minutes.
Clinic In-charge	Review daily dashboard and exceptions.	15 minutes daily.

7.2 Officer workflow

- Morning: review clinics that have not reported, doctor/staff absence flags, previous-day unresolved stock-outs, and high-footfall clinics.
- Afternoon: review referral pendency, disease spike alerts, lab delays, and clinic queue congestion.
- Weekly: compare zone-wise NCD screening, disease trends, medicine consumption, repeat visits, and citizen feedback.
- Monthly: review performance scorecards, support issues, training gaps, and budget/procurement planning.

8. Data Platform, AI, and Analytics Strategy

The data platform is the heart of the proposal. Clinic applications create the data. Dashboards, public-health monitoring, inventory control, and AI alerts convert that data into decisions. The platform should therefore be built with a clean health data model, controlled master data, audit trails, and analytics-ready pipelines.

Analytics area	Indicative metrics / insights
Service utilization	Daily OPD count, hourly footfall, first visit vs repeat visit, age/gender distribution, zone/ward utilization, average time from registration to consultation.
Disease surveillance	Fever trend, cough/respiratory trend, dengue-like symptom flags, gastroenteritis-like symptoms, high-risk clusters, seasonal anomaly alerts.
NCD management	BP screening count, high BP cases, diabetes screening count, high sugar cases, follow-up due, missed follow-ups, repeat visit compliance.
Pharmacy	Stock-out days, near-expiry stock, fast-moving medicines, per-clinic consumption, reorder alerts, supplier/indent status.
Referrals	Referrals raised, referral reason, destination, urgent referrals, closed referrals, pending referrals, long-pending cases.

Clinic operations	Staff attendance, reporting compliance, lab turnaround, queue load, citizen feedback, support tickets.
Financial planning	Medicine demand forecast, device maintenance cost, recurring IT spend, cost per visit estimate, avoidable wastage.

8.1 AI use cases - practical and responsible

AI use case	How it helps	Responsible AI control
Stock-out prediction	Predict medicines likely to run out based on usage, seasonality, and clinic volume.	Recommendations only; final procurement decision by authorized officer.
Disease anomaly detection	Flag unusual rise in fever/respiratory/gastro cases by clinic and ward.	Alert reviewed by public-health officer before action.
NCD follow-up prioritization	Identify patients due for diabetes/hypertension follow-up and prioritize reminders.	Consent-based communication and opt-out controls.
Clinical note assistance	Convert structured selections into clear visit summary and prescription instructions.	Doctor reviews and approves every note.
Citizen chatbot	Answer clinic timing, services, nearest clinic, and general follow-up instructions.	No diagnosis; escalates medical queries to clinic/doctor.
Officer assistant	Summarize weekly trends and generate draft review notes.	Officer validation before circulation.

9. Interoperability with ABDM, eHospital, and Government Systems

The solution should be ABDM-ready and eHospital-aware, but not dependent on all integrations being available on day one. The right strategy is phased interoperability: first standardize clinic data and workflows, then connect to ABHA/ABDM, eHospital-compatible referral systems, and state reporting interfaces as approved by government policy and technical access.

Integration target	Recommended approach
ABHA / ABDM	Support ABHA capture/verification and future consent-based record linking. Use ABDM standards where approved and available.
eHospital@NIC / referral hospitals	Design referral notes and visit summaries in a format that can be exchanged with eHospital or compatible hospital HMIS systems.
State health reporting	Create export templates and dashboards for approved reporting formats.
Laboratory systems	Start with manual result entry; integrate with lab devices or LIS where volume justifies.
Pharmacy procurement systems	Start with stock ledger and indent reports; integrate with e-procurement/ERP later if allowed.
Citizen channels	SMS/WhatsApp/email communications through government-approved gateways and templates.

Important implementation rule

Do not wait for every external integration before starting the pilot. The pilot should prove clinic workflow, clean data capture, dashboards, and inventory/referral value first. ABDM/eHospital integrations can be added in parallel and activated when credentials, approvals, and APIs are available.

10. Architecture and Technical Design

The architecture should be modular, secure, configurable, and deployable on government-preferred infrastructure. It should support central administration while allowing local clinic operations to continue even during short connectivity disruptions.

10.1 Logical architecture

Layer	Components	Key design point
User layer	Reception desk, nurse/triage, doctor console, lab, pharmacy, clinic supervisor, zone officer, commissioner dashboard.	Role-specific interfaces with only necessary fields.
Application layer	Patient registration, visit management, prescriptions, lab, pharmacy, referral, notifications, reports.	Modular services with configurable workflows.
API layer	Secure API gateway, authentication, consent service, integration adapters.	All integrations pass through controlled APIs.
Data layer	Operational database, analytics warehouse/lakehouse, audit logs, backup store.	Separation of operational and analytics workloads.
Analytics layer	Dashboards, alerts, trend analysis, AI models, reporting engine.	Common metric definitions and governed data model.
Security layer	IAM, encryption, audit logging, vulnerability management, monitoring, incident response.	Aligned with government cyber requirements.

10.2 ASCII architecture view



10.3 Suggested technology stack

Component	Suggested choice	Reason
Frontend	Responsive web app and Android tablet mode	Works across reception desktops, doctor laptops, and clinic tablets.
Backend	API-first application services	Clean integration and modular maintenance.
Database	Relational operational database plus analytics warehouse/lakehouse	Reliable transactional records plus scalable analytics.
Dashboards	Power BI / open-source BI / custom dashboard based on government preference	Flexible procurement and hosting options.
AI layer	Python-based models and alert rules, human-approved outputs	K Mati strength in data engineering and Python; avoids unsafe autonomous clinical decisions.

Hosting	Government cloud, India-hosted cloud, or state data center	Supports data sovereignty and government preference.
Security	IAM, encryption, audit logs, backups, WAF, vulnerability scanning	Meets health-data governance needs.

11. Security, Privacy, Compliance, and Data Governance

Health data is highly sensitive. The proposal should include security and privacy controls from the first day rather than treating them as later add-ons.

11.1 Security controls

- Role-based access control for reception, nurse, doctor, pharmacist, lab technician, clinic in-charge, zone officer, health administrator, and system administrator.
- Multi-factor authentication for administrators and officers; strong password policy for clinic roles.
- Encryption in transit using HTTPS/TLS and encryption at rest for databases and backups.
- Audit logs for login, view, create, edit, print, export, prescription, medicine issue, and data download events.
- Data minimization: collect only what is required for clinic care, reporting, and lawful public-health administration.
- Consent-aware communication and record sharing, especially for ABDM-linked data exchange.
- India-hosted logs and backups with retention aligned to government and CERT-In requirements.
- Periodic vulnerability assessment and penetration testing before pilot go-live and before full rollout.
- Incident response playbook with designated point of contact, escalation matrix, and reporting timeline.

11.2 Data governance model

Data domain	Owner	Control
Patient demographics	GBA / BBMP Health Department	Role-based view, field validation, duplicate management.
Clinical visit notes	Authorized clinician / health department	Doctor-authored or doctor-approved records only.
Prescription data	Doctor and pharmacy roles	Doctor approval before issue; stock ledger linkage.
Lab results	Lab role and doctor role	Result entry, verification, and audit history.
Inventory	Clinic pharmacist/storekeeper and zone health officer	Batch/expiry/reorder level controls.
Analytics extracts	Health department data governance committee	Aggregated dashboards by default; identifiable data only where authorized.
AI alerts	Authorized officers and clinicians	Decision support only; no automated clinical diagnosis.

12. Benefits for Public, Clinics, Government, and K Mati

12.1 Benefits for citizens and the public

- Faster registration and lower waiting time through repeat-patient search and token management.
- Clear prescription and visit summary that can be printed or sent digitally where allowed.
- Better continuity of care because previous visits, vitals, medicine, and referrals are not lost in paper registers.
- Improved medicine availability through stock-out alerts and demand planning.
- Better follow-up for chronic conditions such as diabetes and hypertension.
- Easier referral journey with referral note, destination, reason, and follow-up reminder.
- More citizen trust in public clinics through predictable, transparent, and data-supported service delivery.

12.2 Benefits for clinic staff

- Less duplicate register work and easier daily report generation.
- Simple doctor console for common OPD cases and repeat visits.
- Structured triage summary for doctors before consultation.
- Pharmacy stock ledger with low-stock and expiry alerts.
- Referral and follow-up list instead of manual tracking.
- Reduced blame and confusion because actions are logged and visible.

12.3 Benefits for GBA / BBMP and Government

- Near-real-time visibility into clinic utilization, patient categories, disease trends, NCD burden, referrals, stock-outs, and staff reporting.
- Better public-health intelligence for city-level outbreaks and seasonal health patterns.
- Evidence-based procurement and medicine allocation.
- Measurable KPIs for clinic performance and service improvement.
- Improved auditability, transparency, and accountability.
- Foundation for future ABDM integration and urban health command center.
- Better alignment with public primary care, digital health, and citizen-centric governance priorities.

12.4 Benefits for K Mati

- Creates a flagship GovTech and HealthTech reference in Karnataka.
- Builds a reusable platform for other municipal health departments, UPHCs, PHCs, district health programs, and public-health command centers.
- Converts company strengths in data engineering, analytics, AI, dashboards, and healthcare domain exposure into a productized public-sector offer.
- Generates pilot revenue, full rollout project revenue, and recurring managed-service revenue.
- Supports K Mati's strategic shift from generic analytics to measurable outcome partnership.

13. Implementation Plan and Timeline

The project should be implemented in phases so that officials can see value early, frontline users can adapt, and technical integration risk is controlled.

Phase	Timeline	Scope	Key outputs
Phase 0: Discovery and DPR	4 weeks	Workflow study, stakeholder interviews, data model, architecture, pilot clinic selection, implementation plan.	Final DPR, pilot scope, user journeys, budget confirmation, risk plan.
Phase 1: Prototype	4 weeks	Registration, queue, triage, doctor console, pharmacy ledger, basic dashboard.	Clickable prototype and test environment.
Phase 2: Pilot rollout	8 to 10 weeks	20 clinics across representative areas; train users; operate support desk; capture feedback.	Live pilot, adoption dashboard, issue register, pilot performance report.
Phase 3: Hardening	4 weeks	Improve UI, fix field issues, refine reports, security testing, performance testing, data-quality cleanup.	Production-ready platform and rollout pack.
Phase 4: Citywide rollout	4 to 6 months	Wave-wise deployment to all confirmed clinics, with training and support.	All clinics live, officer dashboards active, helpdesk operational.
Phase 5: Optimization	3 months	AI alerts, disease trend analytics, NCD follow-up automation, public-health scorecards.	Advanced analytics and managed operations.

13.1 Rollout wave plan

Wave	Clinics	Selection logic	Expected duration
Pilot wave	20 clinics	Mix of high-volume, medium-volume, low-volume, different zones, different connectivity and staff readiness.	8 to 10 weeks
Wave 1	40 clinics	Clinics with best readiness and high public footfall.	4 to 6 weeks
Wave 2	50 clinics	Remaining high and medium volume clinics.	5 to 7 weeks
Wave 3	50 clinics	Clinics requiring more training or minor hardware setup.	5 to 7 weeks
Wave 4	Remaining clinics	Any additional clinics after confirmed official count.	As per count

14. Budget Plan and Commercial Options

Budget assumption

The exact operational count of Namma Clinics under GBA / BBMP must be confirmed before final commercial submission. This proposal uses a 200-clinic recommended planning scenario and also provides 100, 150, and 250 clinic scenarios. All figures are indicative, exclusive of GST, and subject to procurement scope, hardware availability, hosting choice, and integration approvals.

14.1 Recommended budget model - 200 clinic scenario

Cost component	Basis	Estimated amount
Discovery, DPR, solution design, and pilot preparation	4 weeks, stakeholder workshops, forms/workflow mapping, architecture, pilot plan.	Rs. 25 lakh
Core platform configuration and development	Registration, queue, triage, doctor console, pharmacy, lab, referral, dashboard, admin, notification base.	Rs. 2.25 crore
Analytics and command dashboard	Clinic, zone, city, public-health, pharmacy, referral, NCD, performance dashboards.	Rs. 95 lakh
ABDM/eHospital readiness and API layer	ABHA capture support, consent-ready architecture, referral exchange templates, API gateway.	Rs. 75 lakh
Clinic hardware and peripherals	Approx. Rs. 1.25 lakh per clinic for 200 clinics; includes mix of desktop/tablet/printer/barcode/UPS/basic peripherals as required.	Rs. 2.50 crore
Implementation, training, and change management	Approx. Rs. 0.80 lakh per clinic for 200 clinics.	Rs. 1.60 crore
First-year hosting, helpdesk, maintenance, and support	Approx. Rs. 2.20 lakh per clinic per year for 200 clinics.	Rs. 4.40 crore
Security audit, VAPT, backup/DR setup, monitoring	Pre-go-live testing, monitoring setup, audit readiness.	Rs. 55 lakh
Program management and quality assurance	PMO, documentation, acceptance testing, issue governance.	Rs. 40 lakh
Contingency	Approximately 10 percent reserve for scope	Rs. 1.15 crore

	and field variation.	
Total indicative year-one budget	Recommended 200 clinic citywide scenario.	Rs. 12.80 crore

14.2 Pilot budget - 20 clinic pilot

Cost component	Estimated amount
Discovery, pilot design, and clinic selection	Rs. 15 lakh
Pilot platform configuration and prototype	Rs. 45 lakh
Devices/peripherals for 20 clinics	Rs. 25 lakh
Training and implementation for 20 clinics	Rs. 16 lakh
Pilot hosting, support, helpdesk, and monitoring	Rs. 18 lakh
Dashboards, analytics, and pilot evaluation report	Rs. 25 lakh
Security review and contingency	Rs. 21 lakh
Total indicative pilot budget	Rs. 1.65 crore

14.3 Scenario totals

Scenario	Approx. first-year project cost	Approx. annual recurring from year two	Notes
100 clinics	Rs. 7.25 to 8.25 crore	Rs. 3.0 to 3.5 crore	Good if GBA starts with partial jurisdiction rollout.
150 clinics	Rs. 10.0 to 11.2 crore	Rs. 4.2 to 4.7 crore	Balanced phased city rollout.
200 clinics	Rs. 12.8 to 13.5 crore	Rs. 5.4 to 5.8 crore	Recommended planning scenario.
250 clinics	Rs. 15.6 to 16.8 crore	Rs. 6.5 to 7.2 crore	Use if all current and planned clinics are included.

14.4 Commercial options

Option	Scope	Best for	Indicative commercial model
Option A: Lite digitization	Registration, queue, doctor notes, pharmacy ledger, basic dashboard.	Fast start where budget is limited and existing hardware is available.	Lower upfront cost; fewer integrations; support subscription.
Option B: Recommended platform	All core modules, dashboards, first-year support, hardware bundle, security, and integration readiness.	Citywide service modernization with measurable governance outcomes.	Project implementation plus annual managed support.
Option C: Advanced command center	Recommended platform plus AI alerts, disease surveillance, telemedicine, chatbot, larger command-center dashboard, advanced analytics.	GBA-level public-health intelligence and service transformation.	Higher upfront implementation plus premium annual managed services.

14.5 Payment milestone suggestion

Milestone	Payment share
Work order and project kickoff	10 percent
Discovery and DPR approval	10 percent
Prototype sign-off	15 percent
Pilot go-live	20 percent
Pilot acceptance and hardening closure	15 percent

Wave rollout completion	20 percent
Final acceptance and documentation handover	10 percent

15. Pilot Plan, Success Metrics, and Acceptance Criteria

The pilot should be designed to answer one question: can the platform improve daily clinic operations and officer visibility without overloading clinic staff?

15.1 Pilot clinic selection

- Include high-volume, medium-volume, and low-volume clinics.
- Include clinics from different zones / city corporations under GBA structure.
- Include at least two clinics with weak connectivity to test offline-safe design.
- Include clinics with different physical layouts and patient mix.
- Include clinic staff who are both digitally comfortable and digitally hesitant, so training design is realistic.

15.2 Pilot KPIs

KPI	Pilot target
Daily digital registration compliance	At least 90 percent of OPD visits registered digitally in live clinics by week 6.
Repeat patient retrieval	At least 70 percent repeat patients retrieved without creating duplicate record.
Prescription digital capture	At least 80 percent of consultations with digital prescription or visit summary.
Medicine stock ledger usage	Daily closing stock entered for at least 90 percent pilot clinics.
Referral tracking	All referrals generated digitally with pending/closed status.
Dashboard use	Zone officer dashboard reviewed at least twice weekly during pilot.
User satisfaction	At least 75 percent staff report workflow is usable after training and stabilization.
Support SLA	Critical issues responded to within 2 hours during pilot working window.

15.3 Acceptance criteria

- Pilot clinics complete live operations for at least 30 continuous working days.
- All critical modules work: registration, triage, doctor consultation, prescription, pharmacy, referral, dashboard, and reports.
- Data quality checks show acceptable duplicate, missing field, and synchronization rates.
- Security audit issues categorized as high/critical are fixed before scale rollout.
- Pilot report includes before/after metrics, user feedback, issues, and full rollout recommendations.

16. Operations, Training, Helpdesk, and SLA

A government health platform succeeds only when day-to-day support is reliable. K Mati should propose a managed operations model with defined support roles, issue resolution timelines, training refreshers, and monthly governance reviews.

16.1 Training plan

Audience	Training content	Duration
Reception staff	Patient search, registration, token, duplicate prevention, basic troubleshooting.	Half day

Nurse / triage	Vitals, symptoms, red flags, doctor queue, basic screening forms.	Half day
Doctors	Consultation, diagnosis, prescription, orders, referral, visit summary.	Half day to one day
Pharmacy / store	Medicine master, stock receipt, issue, return, expiry, reorder reports.	Half day
Lab technician	Test order, result entry, verification, reports.	Half day
Clinic in-charge	Daily dashboard, attendance/reporting, issue escalation, inventory review.	Half day
Zone and city officers	Dashboards, alerts, trend review, drill-downs, governance reports.	Half day
Admin / IT team	User management, access, configuration, backups, support processes.	One day

16.2 SLA proposal

Severity	Example	Response target	Resolution / workaround target
Critical	System down for multiple clinics, login outage, data loss risk.	30 minutes	4 hours workaround / 24 hours permanent fix
High	Module failure in one or more clinics, prescription or pharmacy workflow blocked.	2 hours	1 business day
Medium	Report issue, minor workflow bug, non-critical integration issue.	1 business day	3 business days
Low	UI request, enhancement, training clarification.	2 business days	Next planned release or agreed schedule

17. Risk Register and Mitigation Plan

Risk	Impact	Mitigation
Frontline adoption is low	System becomes another reporting burden.	Co-design workflows, use simple screens, train role-wise, identify clinic champions, run field support during first weeks.
Internet connectivity is unstable	Clinic work is interrupted.	Offline-safe registration and local cache for short outages; sync retries; connectivity readiness checklist.
ABDM/eHospital integration delays	Interoperability timeline extends.	Do not block pilot; start with structured export/import and activate APIs when approved.
Data privacy concerns	Public trust and compliance risk.	Data minimization, consent-aware sharing, access control, audit logs, encryption, privacy notices.
Medicine master quality is poor	Inventory reporting becomes unreliable.	Standardized medicine master, batch/expiry fields, pharmacist validation, periodic stock audit.
Doctor time burden increases	Doctors resist use.	Templates, quick-pick fields, minimal mandatory fields, voice-to-text pilot only after clinical approval.
Scope creep	Budget and timeline expand.	Phase-wise scope, change-control committee, documented acceptance criteria.
Cyber incident	Service and trust impact.	CERT-In aligned response plan, log retention, monitoring, backup, VAPT, PoC designation.

18. Governance Model and Decision Structure

The project should have a formal governance mechanism so that clinical, administrative, technical, and procurement decisions are resolved quickly.

Governance body	Participants	Responsibilities	Frequency
Steering Committee	Special Commissioner / senior health officials, K Mati leadership, program director.	Policy decisions, budget, scope, milestones, risk escalation.	Monthly
Project Management Office	GBA/BBMP nodal officer, K Mati PM, technical lead, operations lead.	Daily execution, issue tracking, schedule, dependencies.	Weekly
Clinical Workflow Group	Doctors, nurses, pharmacists, lab technicians, health officers.	Validate forms, workflow, clinical usability, templates.	Fortnightly during design/pilot
Data and Security Committee	Health IT, security officer, data governance officer, K Mati security lead.	Data standards, access rules, security controls, incident response.	Monthly
Pilot Review Group	Pilot clinic in-charges, zone officers, K Mati field team.	Field issues, training gaps, adoption, process fixes.	Weekly during pilot

19. Procurement Strategy and Funding Models

Government procurement should be structured to reduce lock-in, protect data ownership, and support phased delivery. The proposal can be submitted as a concept note first, followed by a DPR workshop and pilot commercial proposal.

19.1 Funding and procurement models

Model	Description	When suitable
Direct pilot order / limited pilot	Small pilot for selected clinics to validate workflows and KPIs.	Fastest route to demonstrate value.
Open tender / RFP	Competitive procurement for citywide deployment.	Appropriate after pilot scope is validated.
GeM / e-procurement listing	If relevant services and SaaS/support categories are available.	Useful for transparent procurement and rate discovery.
PPP / managed service	K Mati operates platform and support as annual managed service under government data ownership.	Useful if GBA wants lower upfront IT management burden.
CSR-supported modules	Partner-funded devices, citizen communication, screening analytics, or digital displays.	Useful for public-health outreach and clinic equipment.
Government budget + phased AMC	Capital implementation plus annual operations and maintenance.	Recommended for long-term public-sector control.

19.2 Contractual safeguards

- Government owns all patient and operational data.
- Data export must be available in documented formats at any time.
- Custom source code escrow or handover terms can be defined for government-funded custom modules.
- Open APIs and standard data dictionary to reduce vendor lock-in.
- Clear service-level commitments and penalties for chronic non-performance.
- Mandatory privacy, security, and audit clauses.
- Exit plan with data transfer, knowledge transfer, and transition support.

20. Appendices

Appendix A: Detailed module matrix

Module	Users	Key outputs
Registration	Reception, data entry, clinic in-charge	Patient ID, daily token, visit creation, patient demographics.
Triage	Nurse, doctor	Vitals, chief complaint, risk flags, doctor queue.
Doctor console	Doctor	Clinical note, diagnosis, prescription, tests, referral, follow-up.
Lab	Lab technician, doctor	Test order, result, verified report.
Pharmacy	Pharmacist/storekeeper, clinic in-charge	Medicine issue, closing stock, stock alerts, expiry report.
Referral	Doctor, referral desk, zone officer	Referral slip, pending/closed status, follow-up list.
Dashboard	Clinic in-charge, zone officer, special commissioner	Operational, public-health, and management insights.
Admin	System administrator, health IT	Users, roles, clinic master, medicine master, configuration.

Appendix B: Role-based access matrix

Role	Can view	Can create/edit	Cannot access
Reception	Patient demographics, queue status	Registration, visit token	Clinical notes beyond necessary routing.
Nurse	Patient profile, current visit, vitals history	Vitals, triage, screening	System admin, citywide reports.
Doctor	Patient history, current visit, lab results, prescriptions	Clinical notes, diagnosis, prescription, referral	Administrative user management.
Pharmacy	Prescription and stock ledger	Medicine issue, stock receipt, stock adjustment with approval	Full clinical notes unless needed for dispensing.
Lab	Test orders and patient identifiers required for test	Result entry and verification	Pharmacy stock and unrelated clinical notes.
Clinic in-charge	Clinic dashboard, reports, staff and stock summary	Operational corrections with audit	Citywide identifiable patient data.
Zone officer	Aggregated and exception data across zone	Comments, approvals, escalations	Unnecessary clinical details.
GBA / Health leadership	Aggregated dashboards, trends, exceptions	Policy-level comments and approvals	Patient-identifiable data unless authorized by policy.

Appendix C: Suggested dashboard tiles

Dashboard	Recommended tiles
Clinic dashboard	Today OPD, waiting queue, average wait, doctor availability, medicine low stock, pending referrals, lab pending, feedback count.
Zone dashboard	Clinic utilization, reporting compliance, stock-outs, disease trends, referrals, high-risk NCD cohort, support tickets.
Commissioner dashboard	City health overview, zone comparison, disease signals, medicine planning, service KPIs, exception map, SLA violations.
Pharmacy dashboard	Fast-moving medicines, stock-out days, expiry risk, indent status, stock transfer needs.

Public-health dashboard	Fever/respiratory/gastro trends, NCD screening, maternal/child referrals, ward heatmap, seasonal alerts.
-------------------------	--

Appendix D: Data dictionary outline

Entity	Important fields
Patient	Patient ID, ABHA optional, name, age/DOB, gender, phone, address, ward, guardian, consent flags.
Visit	Visit ID, date/time, clinic ID, token, visit type, chief complaint, status, doctor ID.
Vitals	BP, pulse, temperature, SpO2, height, weight, BMI, blood sugar, remarks.
Clinical note	Diagnosis, advice, prescription, tests ordered, referral, follow-up date.
Medicine issue	Medicine ID, batch, quantity issued, dosage, days, stock balance.
Lab result	Test ID, sample date, result, unit, interpretation flag, verified by.
Referral	Referral ID, destination, reason, urgency, status, closure date.
Audit log	User, action, timestamp, clinic, record, IP/device, before/after reference.

Appendix E: Draft covering letter

To
 Sri Venkata Rao Chalapathi, IAS
 Special Commissioner
 Greater Bengaluru Authority / BBMP Health Department
 Karnataka

Subject: Proposal for Namma Clinic Digital Health and Operations Platform

Respected Sir,

Kushagramati Analytics Pvt Ltd (K Mati) is pleased to submit this proposal for a unified Namma Clinic Digital Health and Operations Platform. The proposed solution is designed to strengthen citizen access, improve clinic workflows, support digital health records, reduce manual reporting, enable medicine inventory visibility, and provide real-time public-health dashboards for GBA / BBMP leadership.

We request an opportunity to present a 20-clinic pilot plan and conduct a short discovery workshop with the health department, clinic staff, and technology stakeholders. The proposed pilot can be completed in 12 to 14 weeks after approval and can generate measurable evidence for citywide rollout.

We look forward to working with GBA / BBMP to make Namma Clinics more digital, citizen-friendly, measurable, and future-ready.

Yours sincerely,
 Authorized Signatory
 Kushagramati Analytics Pvt Ltd (K Mati)

Appendix F: Source and reference basis

This proposal is based on the following sources and planning assumptions. Final proposal dispatch should verify current official clinic count, official addressee details, GBA/BBMP department structure, current systems in use, procurement route, and integration permissions.

- **Kushagramati Corporate Deck May 22 2026:** Internal company document provided by user.

- **Kushagramati 2.0 Order Acquisition Plan:** Internal company document provided by user.
- **Kushagramati Analytics Strategic Diagnostic and Growth Report:** Internal company document provided by user.
- **eHospital@NIC official website:** <https://ehospital.gov.in/>
- **eHospital@NIC module list:** <https://ehospital.gov.in/ehospitalssso/ehosModuleList.jsp>
- **Ayushman Bharat Digital Mission official website:** <https://abdm.gov.in/>
- **National Health Policy 2017, Ministry of Health and Family Welfare:** <https://mohfw.gov.in/sites/default/files/9147562941489753121.pdf>
- **CERT-In Directions under Section 70B, 28 April 2022:** https://www.cert-in.org.in/PDF/CERT-In_Directions_70B_28.04.2022.pdf
- **Namma Clinic operational context from public reporting:** Times of India report on Namma Clinics requiring waiting area, examination, drug store, laboratory, restrooms and staff room; doctor, nurse, lab technician and health inspecting officer staffing.
- **GBA administrative context from public reporting:** Times of India report on GBA divisions and sub-divisions for administrative functioning.

Appendix G: Immediate next steps for K Mati

21. Confirm current Namma Clinic count, zones, existing IT devices, and current reporting process.
22. Create 10-slide executive deck from this report for first official meeting.
23. Prepare clickable prototype with registration, triage, doctor console, pharmacy ledger, and dashboard screens.
24. Prepare pilot commercial note for 20 clinics with exact inclusions/exclusions.
25. Prepare data privacy and security annexure for government review.
26. Identify K Mati project director, clinical workflow advisor, solution architect, data lead, and field implementation lead.
27. Request a 60-minute presentation meeting and a 2-hour workflow discovery workshop with GBA / BBMP health officials.